



Hydraulic & Engineering Instruments

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OPERATING MANUAL



An ISO 9001 : 2015 Company



**OPERATING INSTRUCTIONS
FOR
V-FUNNEL TEST FINAL**

Hydraulic & Engineering Instruments

(An ISO 9001:2000 Company)

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ISO 9001:2000

V funnel test and V funnel test at T 5 minutes on Self Compacting Concrete

The equipment for V funnel test on self compacting concrete consists of a v shaped funnel used to determine the filling ability (flowability) of the concrete with a maximum aggregate size of 20mm.

The funnel is filled with about 12 liter of concrete and the time taken for it to flow through the apparatus measured. After this the funnel can be refilled concrete and left for 5 minutes to settle. If the concrete shows segregation then the flow time will increase significantly.

Assessment of test

Though the test is designed to measure flowability, The inverted cone shape will cause any liability of the concrete to block to be reflected in the result if, for example there is too much coarse aggregate.

High flow time can also be associated with low deformability due to a high paste viscosity, and with high inter-particle friction. While the apparatus is simple, the effect of the angle of the funnel and the wall effect on the flow of concrete is not clear.

Procedure flow time:

About 12 liter of concrete is needed to perform the test, sampled normally. Set the V-funnel on firm ground. Moisten the inside surface of the funnel. Keep the trapdoor to allow any surplus water to drain. Close the trap door and place a bucket underneath.

Fill the apparatus completely with the concrete without compacting or tamping; simply strike off the concrete level with the top with the trowel.

Open within 10 sec after filling the trap door and allow the concrete to flow out under gravity. Start the stopwatch when the trap door is opened, and record the time for the complete discharge (the flow time). This is taken to be when light is seen from above through the funnel. The whole test has to be performed within 5 minutes.

Procedure flow time at T5 minutes:

Do not clean or moisten the inside surface of the funnel again. Close the trapdoor and refill the V-funnel immediately after measuring the flow time. Place a bucket underneath.

Fill the apparatus completely with concrete without compacting or tapping, simply strike off the concrete level with the top with the trowel.

Open the trapdoor 5 minutes after the second fill of the funnel and allow the concrete to flow out under gravity.

Simultaneously start the stopwatch when the trap door is opened and record the time discharge to complete flow (the flow time at T5 minutes). This is to be taken when light is seen from above through the funnel.

Interpretation of result:

This test measures the ease of flow of concrete, shorter flow time indicates greater flow ability. For SCC a flow time of 10 seconds is considered appropriate.

The inverted cone shape restricts the flow, and prolonged flow times may give some indication of the susceptibility of the mix to blocking. After 5 minutes of settling, segregation of concrete will show a less continuous flow with an increase in flow time.